BEHAVIORAL DEDUPLICATION AND DIVERSITY-GUIDED SELECTION FOR SAMPLE-EFFICIENT FUNSEARCH

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ABSTRACT

We propose **behavioral-fingerprint deduplication** for FunSearch, an LLM-driven evolutionary program search framework. LLM proposers repeatedly produce functionally equivalent heuristics: on an unmodified run we observe 45% of proposals scoring identically to an ancestor—a conservative proxy for behavioural redundancy whose sandbox evaluations carry no new information (and, under bounded budgets, also waste future API quota). Our method executes each candidate on ten lightweight probing instances, hashes the resulting 375-dimensional decision sequence, and rejects exact matches before full evaluation; we further integrate diversity-guided sampling and ReEvo reflective evolution (Ye et al., 2024) into the same pipeline. On OR3 online bin packing under a 150-sample budget, deduplication alone reduces *samples-to-best* by 38% ($146 \rightarrow 90$) without degrading the best score. An offline **Behavioural Coverage Rate (BCR)** analysis (n=1 per condition, exploratory) suggests that deduplication alone may reach broader behavioural coverage than deduplication combined with diversity-guided sampling—a hypothesis flagged for multi-seed follow-up. Adding ReEvo on top of deduplication and diversity-guided sampling yields the lowest-variance condition in our runs (mean -209.66, std 0.59, n=4), consistent with the landscape-smoothing mechanism reported by Ye et al. (2024).[1]

1 INTRODUCTION

FunSearch (Romera-Paredes et al., 2024) evolves programs by iterating an LLM proposer against an evaluator: each round, the LLM drafts a new heuristic given recent high-scoring ancestors, and a sandbox measures the candidate's score. The paradigm has recently been scaled up to industrial settings — AlphaEvolve (Novikov et al., 2025) reports substantial improvements on Google Borg datacenter scheduling using this same LLM-evolve loop — and online bin packing has emerged as one of its canonical benchmarks (Sim et al., 2025; Liu et al., 2024). Under limited API budgets, however, the *sample efficiency* of the proposer becomes the central differentiator (Ye et al., 2024): the question is not whether a reasonable heuristic can be found, but how many LLM calls it takes.

Our preliminary baseline exposes a concrete waste source. Running unmodified FunSearch on the OR3 instance set for 53 samples, 45% of all programs produced scores *identical* to programs already in the archive—a conservative proxy for behavioural redundancy (same score need not imply same per-item decision sequence, but the converse holds) that syntactic hashing does not catch, and which the 375-dimensional fingerprint introduced in §2 verifies directly. This matches the open direction identified by van Stein et al. (2025), who argue that niching and portfolio-style behavioural analysis are under-explored in LLM-driven meta-heuristics. Existing FunSearch variants do not deduplicate at the behavioural level and do not explicitly constrain diversity of accepted programs.

We report three results, all under a 150-sample budget. **(Main)** A three-level *behavioural-fingerprint funnel* (AST-hash $\rightarrow$ probing-fingerprint match $\rightarrow$ a disabled cosine layer), inspired by ReEvo's F-value deduplication (Ye et al., 2024) but lifted to the behaviour level; on OR3 under a 150-sample

[1]Reproducibility: https://colab.research.google.com/github/jaycee6666/efficient-funsearch/blob/main/notebooks/efficient_funsearch_colab.ipynb.

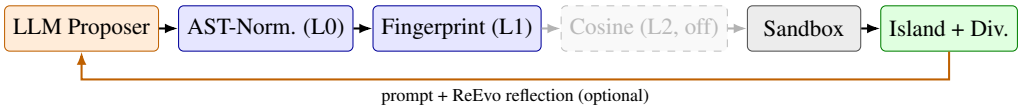

Figure 1: System pipeline. Candidates are normalised (L0) and fingerprinted (L1) before sandbox evaluation; cosine (L2) is retained in code but disabled at run time (Section 2.2). Survivors enter an island archive with diversity-guided sampling; archived programs feed the LLM via prompt construction and, optionally, ReEvo reflection.

budget it reduces *samples-to-best* by 38% without degrading best score. **(Extension)** We integrate diversity-guided cluster sampling ($\mathrm{Score}(c) = \mathrm{Perf}(c) + \beta \cdot \mathrm{Div}(c)$, §2.3) and ReEvo reflective evolution on top of deduplication inside the same pipeline and analyse their stacked behaviour; reflection-only is not separately ablated (§6 #4). **(Tool)** We introduce an offline **Behavioural Coverage Rate** (BCR) metric inspired by MAP-Elites (Mouret & Clune, 2015); on an exploratory pair of runs (n=1 per condition) it suggests that deduplication alone may reach full behavioural coverage faster than deduplication combined with diversity-guided sampling, consistent with the view that at the behaviour level *removing redundancy* and *promoting novelty* may not be simply additive (van Stein et al., 2025).

2 Method

2.1 Pipeline Overview

The original FunSearch pipeline alternates between an LLM sampler and a sandboxed evaluator, with island-based selection maintaining diversity at the population level (Romera-Paredes et al., 2024). Our enhanced pipeline inserts a deduplication funnel *before* the expensive sandbox, and optionally wires in diversity-guided sampling and ReEvo reflection around the database (Figure 1). Only programs surviving all enabled dedup levels incur a full sandbox evaluation; accepted programs additionally trigger a lightweight LLM reflection call whose output is stored and replayed as prompt context in later rounds.

2.2 Behavioral Fingerprint

Probing instances. We hand-designed 10 bin-packing probes spanning uniform-medium, large-item, small-varied, bimodal, descending-spread, ascending-spread, sawtooth, tight-pairs, wide-capacity, and an OR3-derived sample (Appendix B). Capacity is drawn from $\{100, 150, 200\}$ and items from $[20, 140]$ to match the OR3 distribution; each probe contains 35–40 items, for a total of 375 decisions.

Fingerprint definition. For a candidate priority function f and probe p_i, we simulate the online first-fit-by-priority assignment: at each step the bin with the highest $f(item, bins)$ is chosen (opening a new bin when needed). The k-th decision is the index of the chosen bin. Concatenating the decisions across all ten probes yields a deterministic 375-dimensional integer tuple, the program's *behavioural fingerprint*.

Three-level funnel. Level 0 normalises the candidate source (variable renaming, comment stripping) through an AST rewriter and hashes the result with SHA-256. Level 1 hashes the fingerprint tuple and performs an $O(1)$ exact-match lookup against an archive. On the proxy probe set, Level 1 matches are exact by construction; at the *fingerprint-match stage* this produces no threshold-based false positive (we discuss the generalisation of this equivalence to out-of-distribution inputs in §4 and §6 #2). Level 2 was implemented as a cosine-similarity check with threshold 0.98; a calibration on 53 baseline programs showed that 98.5% of genuinely different programs would be flagged duplicate, so it is disabled and kept only as a negative result (§4, "Cosine on discrete decision vectors is degenerate").

Overhead. Normalisation is < 1 ms, executing one probe < 3 ms, the overall fingerprint < 50 ms, and the hash lookup negligible — all dominated by the ~ 11 s sandbox evaluation they replace.

2.3 DIVERSITY-GUIDED CLUSTER SAMPLING

When `DIVERSITY_ENABLED=1`, the prompt-construction step inside each island's `get_prompt()` (in `programs_database.py`) selects k=2 clusters via softmax over a *combined* score

$$\mathrm{Score}(c) \;=\; \widehat{\mathrm{Perf}}(c) \;+\; \beta \cdot \widehat{\mathrm{Div}}(c),$$

with both terms min-max normalised across the island's clusters and $\mathrm{Div}(c) = 1 - \overline{\cos}(\tilde{\mathbf{s}}_c, \tilde{\mathbf{s}}_{c'})_{c' \neq c}$ the row-centred mean cosine distance from c to the other clusters; here $\mathbf{s}_c \in \mathbb{R}^{|\mathcal{D}|}$ ($|\mathcal{D}|$=20) is the per-OR3-file score signature already used as the archive's bucket key (so Div is defined on score patterns, not on the 375-dim probing fingerprint). Row-centring decouples it from absolute level, which $\widehat{\mathrm{Perf}}$ already captures. The exploration weight β decays linearly from β_0=0.3 to 0 over the first 350 accepted programs, so early sampling prefers behaviourally under-represented clusters and late sampling reverts to pure performance-driven exploitation. This is the mechanism activated in our *dedup+div* and *reevo* runs.

2.4 REEVO REFLECTIVE EVOLUTION

Following Ye et al. (2024), we add a per-accepted-program reflection step: after a program is scored, the LLM is asked in 1–2 sentences to describe the algorithmic strategy the program uses and how it differs from a simple greedy approach. We keep the top-3 highest-scoring reflections and inject them as comments into the next prompt as "strategy insights from top-performing programs" (full prompt in Appendix C). Context: ReEvo's landscape-smoothing analysis (Ye et al., 2024, Table 4) reports a $4.5\times$ increase in correlation length and 45.8% objective improvement on related heuristic-design benchmarks, which provides a plausible mechanism consistent with the variance reduction we observe (§3.2).

2.5 OFFLINE BEHAVIOURAL COVERAGE RATE (BCR)

BCR is a post-hoc diagnostic that characterises how quickly a run visits distinct behavioural regions. It is defined only for fingerprint-exporting conditions (*dedup*, *dedup+div*); baseline bypasses the filter, and our ReEvo run did not set `EXPORT_FINGERPRINTS=1`. Given the union of fingerprints, we fit a single k-means (K=8) once, then replay each condition's time series: $\mathrm{BCR}(t) = |\{c(\mathbf{f}_s) : s \leq t\}|/K$. Offline fit-then-replay avoids cluster-relabelling drift at this sample size. The coverage intuition follows MAP-Elites (Mouret & Clune, 2015; Novikov et al., 2025); we do not claim alignment with van Stein et al.'s runtime-trajectory definition (van Stein et al., 2025).

3 EXPERIMENTS

3.1 SETUP

Task. Online bin packing (OBP) receives items $s_1, s_2, \ldots$ one at a time and must assign each to a bin irrevocably on arrival; the heuristic h scores every open bin and places the item in the highest-scoring feasible one (or opens a new bin). The score reported in this paper is $-\frac{1}{|\mathcal{D}|}\sum_{d\in\mathcal{D}} \mathrm{bins_used}(h, d)$, i.e., the negated mean number of bins used over $|\mathcal{D}|$=20 OR3 files (`u500_00`–`u500_19`, each 500 items, capacity 150); higher is better. The L1 lower bound $\lceil \sum s_i/C \rceil$ averages to -201.2 on OR3, so a score of -209.66 corresponds to $\sim$ 8.5 bins above the theoretical minimum. All experiments evaluate on the OR3 dataset only; Weibull-5k from the original FunSearch protocol (Romera-Paredes et al., 2024) is not used in this report. The LLM proposer is `gpt-5-nano` via the course API, with 10 islands, 4 samples-per-prompt, and a 30 s sandbox timeout. Each run is capped at 150 LLM samples — well below the 20,000-call budget — to keep the comparison within the "low-sample regime" studied by Ye et al. (2024). We evaluate four conditions: *baseline* (unmodified FunSearch, n=2), *dedup* (fingerprint funnel only, n=4), *dedup+div* (funnel plus diversity-guided sampling, n=4), and *reevo* (funnel + diversity-guided sampling + ReEvo reflection stacked, n=4; reflection-only is not separately ablated—see §6 #4). Across all runs the proposer issues $\sim$ 2,100 samples; the four *reevo* runs add up to one reflection call per accepted program.

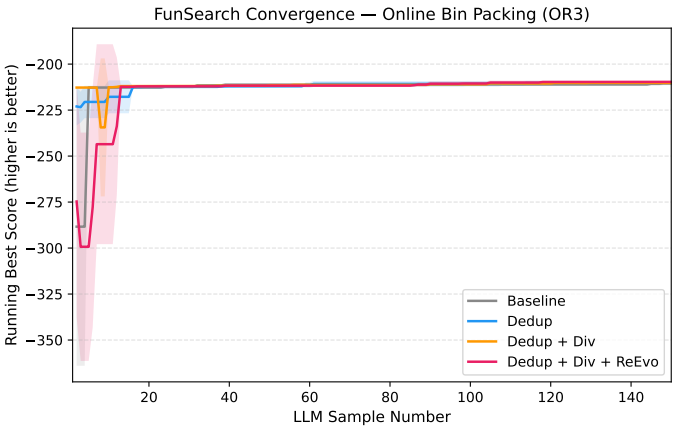

Figure 2: Best-score convergence versus sample number across the four conditions (mean $\pm$ std band, n=2–4). All conditions land within -210 ± 1; the ReEvo condition is the only one whose mean is above -210.

Table 1: Main results at a 150-sample budget. *Samples-to-best* is the first sample number at which a run attains its best score; *Dup rate* is the fraction of candidates rejected by the dedup filter (N/A for baseline, which bypasses the filter).

Condition	n	Best score (mean $\pm$ std)	Samples-to-best	Dup rate
Baseline	2	-210.52 ± 1.31	146 ± 2	N/A
Dedup	4	-210.04 ± 1.00	$\mathbf{90 \pm 36}$	0.401 ± 0.046
Dedup+Div	4	-210.20 ± 0.68	108 ± 21	0.342 ± 0.067
ReEvo	4	$\mathbf{-209.66 \pm 0.59}$	101 ± 16	0.314 ± 0.034

3.2 Main Results

Table 1 and Figure 2 reveal four observations.

(O1) Main contribution. Baseline→Dedup reduces samples-to-best from 146 to 90, a 38.4% **reduction** in the number of LLM samples the proposer consumes before the run's best program is discovered. The best score itself is unchanged ($-210.52 \rightarrow -210.04$).

(O2) No observable performance degradation. All four conditions have best scores in -210 ± 1. At this budget and sample size (n=2–4) we observe no measurable best-score cost from deduplication, diversity-guided sampling, or reflection — an observation consistent with (but, per §4, "Cosine on discrete decision vectors is degenerate", not a direct verification of) a low false-positive rate on the full input distribution.

(O3) Full stack yields the lowest variance. The *reevo* condition (dedup + diversity + reflection) is the only one whose mean is above -210 (-209.66 ± 0.59). This is qualitatively consistent with the landscape-smoothing effect reported for reflection by Ye et al. (2024), but because reflection is stacked on top of dedup and diversity-guided sampling in these runs, we cannot separately attribute the improvement to reflection; n=4 is also insufficient for significance testing.

(O4) Dup rates decrease as extra signals are added. Among dedup-equipped conditions, the rejection rate drops from 0.401 (dedup) to 0.342 (dedup+div) to 0.314 (reevo). Because the archive distribution and prompt context co-vary across these conditions (stronger signals also change which programs survive into the archive), we read this as a correlational observation rather than evidence that diversity or reflection causally pushes the LLM toward new behaviours. (We do not include baseline in this trend because it does not route candidates through the dedup filter; the 45% natural duplicate rate mentioned in §1 is computed independently on identical-score counts.)

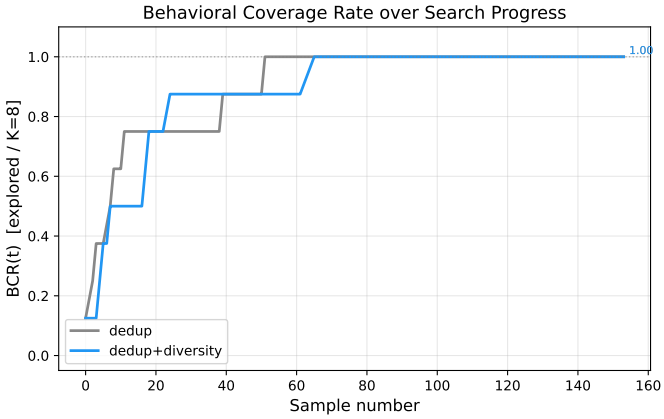

Figure 3: Behavioural Coverage Rate over sample index (exploratory; n=1 per condition). Dedup-only (blue) reaches full coverage (BCR$=1$) at sample 51; Dedup+Div (orange) reaches full coverage at sample 65. Only these two conditions export fingerprints in the current run set.

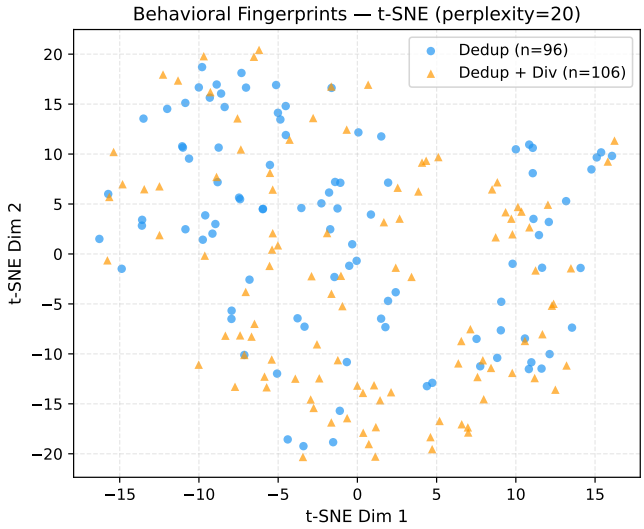

Figure 4: t-SNE ($perplexity$=20) of $\sim$ 200 fingerprint vectors (each 375-dim) collected from the fingerprint-exporting *Dedup* and *Dedup+Div* runs. The two point clouds overlap substantially but Dedup+Div spreads out more within each cluster.

3.3 Behavioural Coverage Analysis

Figure 3 plots BCR(t) for the two fingerprint-exporting conditions (n=1 each; this is exploratory rather than a mechanism claim): Dedup-only reaches $\mathrm{BCR} = 1.0$ at sample 51, whereas Dedup+Div takes until sample 65. **Counterintuitively,** under this single pair of runs, diversity-guided sampling appears to *slow* cluster visitation. A plausible reading is that once duplicates are filtered, retained samples are already diverse by construction, so the extra diversity constraint concentrates exploration within already-reached clusters rather than reaching new ones; confirming this would require multi-seed replication at the same n used for the main table (van Stein et al., 2025).

3.4 FINGERPRINT SPACE VISUALISATION

Figure 4 visualises fingerprints from the two runs with `EXPORT_FINGERPRINTS=1`. Both conditions visit overlapping regions; Dedup+Div spreads further *inside* each cluster while Dedup-only reaches more distinct clusters, consistent with Figure 3.

All four ReEvo runs plateau early and break through at sample 87–123, consistent with the reflection buffer needing time to accumulate high-scoring strategies (per-run curves in Appendix E; $n{=}4$ precludes a causal claim).

4 DISCUSSION

Dedup may subsume first-order diversity. The 38% samples-to-best reduction together with the exploratory BCR curves ($n{=}1$ per condition) are consistent with the hypothesis that removing behavioural redundancy already delivers much of the exploration effect that explicit diversity mechanisms target, and that stacking diversity-guided sampling on top can in fact slow cluster visitation, spreading points within already-reached clusters rather than reaching new ones. This would echo Novelty Search's observation that penalising redundancy can substitute for rewarding novelty directly (Lehman & Stanley, 2011); confirming it requires multi-seed replication.

Cosine on discrete decision vectors is degenerate. Calibration (§2.2) shows 98.5% of distinct-pair similarities > 0.98, so Level 2 is disabled; Hamming/edit-distance or learned metrics are needed.

Reflection-plus-stack consistent with landscape smoothing. The full-stack condition's highest mean (-209.66) and lowest variance (0.59), with the late-breakthrough pattern (Appendix E), are qualitatively consistent with Ye et al. (2024)'s landscape-smoothing mechanism; reflection cannot be isolated here as it runs with dedup and diversity active.

5 RELATED WORK

LLM-driven evolutionary program search. FunSearch (Romera-Paredes et al., 2024) established LLM-evolve for heuristic discovery; AlphaEvolve (Novikov et al., 2025) scaled it to industrial tasks and identifies FunSearch's single-LLM / single-evaluator structure as a limiting factor (Novikov et al., 2025, §5). EoH (Liu et al., 2024) and the online-bin-packing survey (Sim et al., 2025) explore the same space for combinatorial heuristics. We retain FunSearch's minimalist loop and target the proposer's sample efficiency at the behaviour level.

Reflection and behaviour-driven search. ReEvo (Ye et al., 2024) introduces LLM reflection between generations; we integrate it inside the FunSearch loop. BCR is inspired by MAP-Elites (Mouret & Clune, 2015)/Novelty Search (Lehman & Stanley, 2011) but measures cluster visitation over time; van Stein et al. (van Stein et al., 2025) advocate runtime-trajectory behaviour-space analysis. Our probe-decision fingerprint sits between syntactic AST-hash dedup and undecidable semantic equivalence.

6 LIMITATIONS

(1) Statistical power: $n{=}2$–4 (main table), $n{=}1$ (BCR/t-SNE); exploratory, not significance-tested. **(2) Probe generalisation:** fingerprint equivalence is not verified end-to-end on full OR3. **(3) Single** Div **metric:** row-centred cosine only; edit distance, learned embeddings, and per-program filter-level rejection are future work. **(4) Reflection not isolated and probes use `exec()` (5 s timeout) rather than the forked sandbox.**

7 CONCLUSION

Behavioural-fingerprint dedup cuts samples-to-best by 38% on OR3 with no best-score loss; exploratory BCR ($n{=}1$) suggests diversity-guided sampling slows cluster coverage; full-stack ReEvo gives the lowest variance ($n{=}4$).

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A REPRODUCIBILITY AND IMPLEMENTATION NOTES

All results are logged from runs conducted between 2026-04-15 and 2026-04-21 on the course-provided `gpt-5-nano` endpoint. Per-condition raw run logs (`funsearch-baseline/logs/layer*/run_log.csv`) and the aggregation script `scripts/collect_results.py` are included in the released repository; the reported means and standard deviations are reproduced exactly by that script. (The LLM API is non-deterministic, so we do not claim seed-level reproducibility of fresh runs; the released CSV logs are the authoritative artefact.)

API robustness. The proposer LLM wrapper uses a 60 s socket timeout and at most 5 exponential-backoff retries, configurable via `API_TIMEOUT` and `API_MAX_RETRIES`. The ReEvo reflection helper is a separate code path with a fixed 30 s timeout and no retry loop; reflection failures are non-fatal (the run continues without storing that reflection). The proposer bound was added on 2026-04-21 to cap wall-clock on re-runs; it does not affect the logged 150-sample results, which completed under a previous, unbounded retry configuration. The Colab notebook linked in the abstract regenerates every table and figure from the raw CSV logs without rerunning the LLM loop.

B PROBING INSTANCES

Name	Capacity	Items	Description
or3_sample	150	40	First 40 items of OR3 `u500_00`
medium_uniform	150	40	$[30, 31, \ldots, 69]$; pairwise packing possible
large_items	150	35	$[75, 76, \ldots, 109]$; ≤ 2 per bin
small_varied	150	40	Non-monotonic items in $[20, 38]$
bimodal	150	36	18 large (~ 100) + 18 small (~ 25)
descending_spread	150	35	$140, 137, \ldots, 38$ (FFD-style stress)
ascending_spread	150	35	$25, 28, \ldots, 127$
sawtooth	150	36	Periodic $\{30, 50, 70, 90, 110, 130\} \times 6$
tight_pairs	100	40	Items $\sim 50\%$ capacity; many feasible bins
wide_capacity	200	38	Deep bin states, items in $[30, 100]$
Total		375	Fingerprint dimensionality

Table 2: Ten probing instances used to compute behavioural fingerprints. Each instance is deterministic; the fingerprint is the concatenation of the online bin-assignment decisions over all 375 items.

C REEVO REFLECTION PROMPT

After a candidate program is accepted (passes dedup and receives a valid score), we issue the following reflection call to the same LLM backbone:

```
The following Python function is a heuristic for online bin
packing.  It scored {score} (higher is better; optimal is
around −200).
```python
{program body}
```
In 1--2 sentences, describe the algorithmic strategy this
function uses and how it differs from a simple greedy
approach. Be concise.
```

The reflection text (max 150 tokens) is stored by program-body hash. At prompt construction time, the top-3 reflections by score are appended as Python comments under the header `# Strategy insights from top-performing programs:` before the next LLM call.

D HYPERPARAMETERS

Component	Parameter	Value
FunSearch	`num_islands`	10
	`samples_per_prompt`	4
	`functions_per_prompt`	2
	`cluster_sampling_temperature_init`	0.1
	`evaluate_timeout_seconds`	30
LLM	model	`gpt-5-nano`
	`max_tokens`	4096 (main), 150 (reflection)
Dedup	`level0_enabled`	`True`
	`level1_enabled`	`True`
	`level2_enabled`	`False` (see §2.2)
	`cosine_threshold`	0.98
	`probe_timeout_seconds`	5
Diversity	`beta_init`	0.3
	`beta_decay_period`	350
ReEvo	`max_reflections`	20
	top-k for prompt	3

Table 3: Hyperparameter summary. Sources: `implementation/config.py`, `src/dedup/dedup_config.py`, and the ReEvo integration block of `funsearch_bin_packing_llm_api.py`.

E PER-RUN REEVO CONVERGENCE

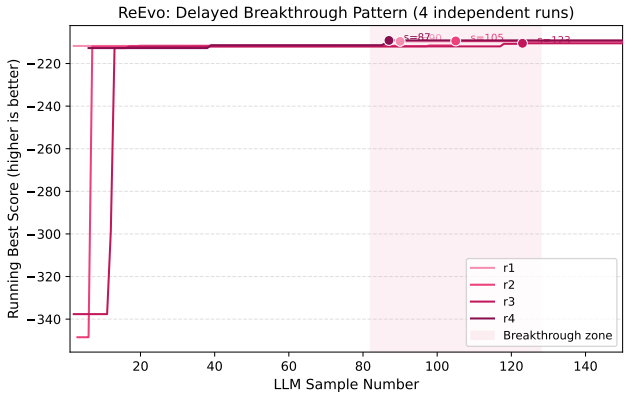

Figure 5: Per-run ReEvo convergence curves (n=4). All four runs show a late improvement at roughly sample 87–123, after an initial plateau.